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INFINIBAND ROUTING TABLE OPTIMIZATIONS FOR SCIENTIFIC APPLICATIONS

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ABSTRACT

The achievable performance on Infiniband networks is governed by the latencies and bandwidths of communication channels as well as by contention within the network. Currently Infiniband uses static routing to transfer messages and thus does not take into account dynamic loading of the channels. By interrogating the network routing tables we quantify the contention that occurs for a number of communication patterns using a large-scale (1024 processor) system. Empirical data confirms our contention calculation almost exactly. Custom routing tables are defined that provide both optimum and worst-case performance for a large-range of communication patterns. Performance differences can be as large as 12× (from optimum to worst-case). Two large-scale applications show a run-time improvement of between 10-20% and up to a 40% improvement in just their communication time when using optimized routing tables. The approach taken is applicable to many Infiniband systems, and we expect the performance improvements to be even greater on larger-scale systems.

Keywords: Large-scale processing, Scientific Applications, Infiniband, Fat-tree Networks, Static Routing

1. Introduction

The performance of an HPC system is determined by the capabilities of the compute nodes, the capabilities of the interconnection network, and the way in which applications exercise these resources. Infiniband can be classed as a commodity-based interconnect that is increasingly used in HPC systems. Though not originally developed for the HPC market, Infiniband networks are found in other areas including telecommunications and storage. The specification of a standard for Infiniband has encouraged many manufacturers to deliver both HCAs (Host Channel Adapters), and switch components. Large-scale HPC systems that use Infiniband

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include the Thunderbird cluster at Sandia National Laboratory [1], System-X at Virginia Tech [2], and Roadrunner at Los Alamos [3]. The performance of an individual communication channel boasts high bandwidth with current networks achieving a 4× DDR (Double Data Rate) performance of 1.6GB/s, and single 0-byte message latency from HCA to HCA of 1.3μs [4]. Infiniband switch chips currently contain a 24-port cross-bar which allows 12-ary fat-trees, as well as other topologies, to be built. The main manufacturer of Infiniband technology, Mellanox, produces cross-bar chips packaged into switch chassis containing a total of 288 ports that are supplied by multiple vendors, such as Voltaire and Silver Storm.

In this work we undertake a performance analysis and optimization of a system that has a single 288-port Infiniband switch. We are not concerned with the peak performance of individual components of the network but rather examine how it performs for particular communication patterns that arise in many large-scale scientific applications. We consider a communication pattern to consist of many communications between distinct processor pairs in the system. The majority of parallel scientific applications correspond to the SPMD (Single Program Multiple Data) programming model in which all application processors perform a computation step at approximately the same time followed by a communication step at the same time. By considering the communication patterns we are effectively examining and optimizing the performance of the communication steps in an application.

The motivation for this work comes from our application performance experience with adaptive and static routed interconnects. Current Infiniband networks use static routing to route messages between source and destination nodes. That is, a deterministic path is always used to transfer messages between one particular node and another. The routing does not adapt with the dynamic loading on the network and contention in particular communication channels can result. Similar routing is used in Myrinet networks. Adaptive routing uses the dynamic loading in the network to determine how each message segment should progress. For instance in the Quadrics QSNet fat-tree networks [5] message segments are adaptively routed when progressing up the tree and statically routed when progressing down the tree (to eliminate dead-lock). The added capabilities within the Quadrics network can alleviate much of the contention issues that arise in statically routed networks and hence provide higher levels of achievable performance for a given channel bandwidth. Adaptive routing is not available on current Infiniband networks.

A further challenge is that compute-nodes within a system are not always configured such that their logical order, as visible to an application, matches the physical ordering on the interconnect switches. If logical neighbors are not physical neighbors, as far as possible, on the switch chips much of the traffic for local neighbor communication will be routed through switches in upper fat-tree levels, thus increasing contention. In other networks the problem is alleviated by excess bandwidth. For example, in the Cray XT3 and XT4 systems the communication channel bandwidth exceeds that that can be injected by a single node and thus reduces contention [6].

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Another challenge that can intensify contention is the increasing number of cores per socket, with multi-core processors even more MPI processes could share the interconnect network.

The rest of the paper is organized as follows. In Section 2 static routing as used within a typical Infiniband network is described along with details on the experimental system used in this work. The characteristics of communication patterns are described in Section 3 along with a model to calculate the contention that arises for a particular pattern given a set of routing tables. In section 4 we analyze the contention that arises in our experimental system and show the result of using optimized routing tables for a range of communication patterns. Empirical data is used to confirm the contention in Section 5. The impact of this optimization on the performance of two applications is quantified in Section 6. Conclusions from this work are discussed in Section 7.

1.1. *Contributions of this work*

We demonstrate that, for a particular application communication pattern, the performance of an Infiniband network is greatly affected by contention within the network. Contention is quantified for a communication pattern by interrogating the routing tables contained within each switch chip in the network. The current emphasis with Infiniband routing is on the algorithms used by the subnet manager to quickly discover the nodes within the network fabric, but little is done after the fabric is operational.

The methods we use to optimize the network performance arise from an application centric viewpoint. Rather than optimizing particular network features such as bandwidth, latency or collectives, we consider the application requirements and customize the fabric accordingly. Optimization was achieved by a strict control of the routing tables to match typically occurring communication patterns. The approach can be extended to include more of a dynamic application-aware routing process. A single optimized routing table is presented that encompasses a range of communication patterns that commonly occur in scientific codes. We envision that such an optimization could be utilized in a large number of systems given the increasing popularity of Infiniband. Work on incorporating optimized routing tables in OpenSM, similar to those presented here, was reported in [7]. Recent novel adaptive routing methods for Fat-trees include [8].

The optimization of a 1024-processor system results with much improved communication times on individual communication patterns, as well as 10-20% improvement in run-time on two large-scale applications. The actual improvement in just the communication component of the application runtime is also examined and shows a 20-40% improvement. Also since contention increases with scale, and cluster sizes are increasing, the results of this work may be more relevant in the future. Earlier work on analyzing the performance of Infiniband networks for scientific applications includes [9] which examined the sensitivity of Infiniband latency and

bandwidth on overall performance. This work extends our earlier analysis reported in [10] on optimizing routing Infiniband routing tables.

2. Infiniband networks for high performance computing

Infiniband is based on an extensive specification [11]. It was designed for enterprise network replacement rather than specifically for HPC. The switching hardware lacks some features that exist in other interconnects, such as hardware collectives and adaptive routing (e.g. Quadrics QsNet [5] and the Blue Gene/L network [12]). Even with these limitations Infiniband fabrics have scaled to thousands of nodes.

A typical Infiniband fat-tree network is constructed from 24-port cross-bar switches in which 12 ports are used as down-links and 12 ports as up-links in one or more switch levels in the network. The down-links in the first level of switches connect to Host Channel Adapters (HCAs) in each of the compute-nodes, and the up-links connect to a higher level of switches. The 24-port switches enable the easy implementation of 12-ary fat-tree networks as well as other reduced connectivity trees [1]. A switch chassis, consisting of a total of 288-ports, is often the network building block in large-scale Infiniband systems.

As an example the internal structure of the Voltaire ISR 9288 switch is shown in Figure 1. Each rectangular block represents an individual 24-port cross-bar. Up to 288 compute-nodes are attached as leaf nodes in the system. The first level of cross-bar switches is configured with 12 down-links, and 12 up-links. The second level is configured with 24 down-links. The ISR 9288 switch can also be used as part of a larger scale system, with for example 144 of the 288 ports connected to nodes and 144 ports connected to a higher level of switches. In this configuration, the ISR 9288 is arranged as three switch-levels each with 12 cross-bars.

The cluster used in this analysis consisted of a single Voltaire ISR 9288 switch interconnecting 257 nodes (256 compute-nodes and a head-node). Each node consists of dual-socket, dual-core AMD Opterons (i.e. four processors per node) operating at 2.0-GHz. Each node has 4-GB of memory and a single Mellanox 4x PCI-Express MT25208 Infinihost III Ex HCA. Renumbering of system nodes took place so as to guarantee that the logical node numbering visible to applications are physical neighbors in the switch fabric. For instance, the first 12 nodes of the system are directly connected to the first cross-bar in the first level of cross-bar switches. This is an important consideration as we will see when we consider the contention that can arise within the network.

Infiniband software consists of a few distinct levels: an IB stack which is used by the host to communicate to the HCA; an MPI stack used by the application to communicate through the fabric; and a Subnet Manager (SM), in our case OpenSM, that initializes the fabric by assigning LIDs (local subnet identifier) and programming the LFT's (LID forwarding tables) for routing messages. There are also GUIDs (global unique identifiers) which are unique across subnets therefore unique for a whole fabric.

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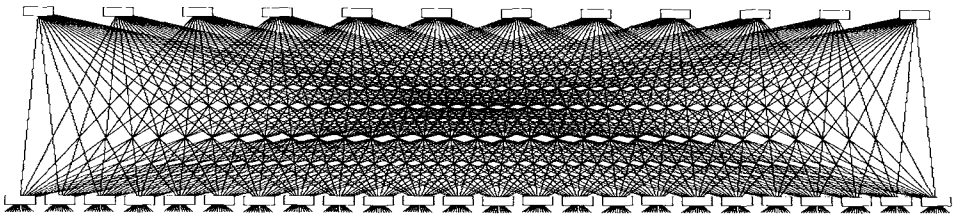

Fig. 1. A 288-node cluster using 2-levels of 24-port cross-bar switches.

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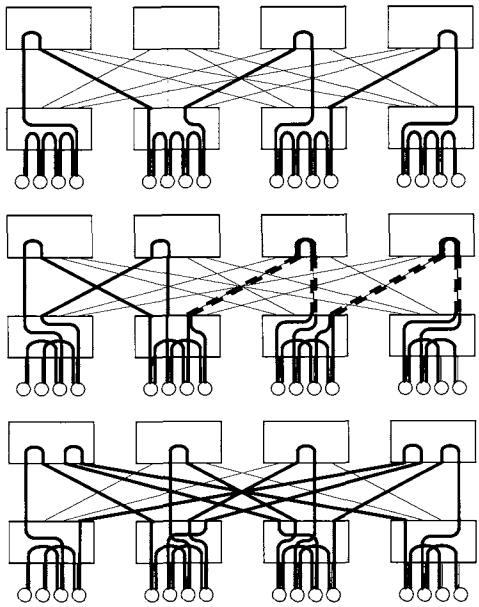
Routing in Infiniband networks is destination-based static routing. When a packet arrives on any cross-bar input port, the output port is determined by looking up the destination LID in the LFT which has one entry per LID in the network. The routing table (or map) for the entire fabric consists of all of the LFTs. In normal operation the SM discovers the network and builds the routing table using one of several methods, for example: minimum hop or UP/DOWN.

Static routing of messages can result in contention in the network when many messages are sent between distinct pairs of nodes at the same time. For instance consider a small 16-port network shown in Figure 2. A two-level quaternary tree is used for illustration purposes. A set of communications is shown by thick curves in Figure 2(a) - in which each node N_i communicates with node $N_i \pm 1$ (nodes are numbered sequentially from left to right). In Figure 2(a) all communications proceed at the same time as no contention occurs - only one message progresses upwards from any first level switch.

A further example set of communications is shown in Figures 2(b) and 2(c) in which each node N_i communicates with node $N_i \pm 2$. The possible contention in the network occurs when two messages, originating from the same first level switch, are routed over the same up-link. This is shown in the example of Figure 2(b) by the thick dashed lines. However, it is possible for all of these communications to progress without contention if the routing table in each switch permits - an example of routing the same set of messages free of contention is shown in Figure 2(c).

The definition of routing tables that provide contention free communications for particular communication patterns is the subject of this work and described further in Section 4. It builds upon knowledge of commonly occurring communication patterns in scientific codes as described in Section 3.

It should be noted that the Infiniband standard also provides for Multi-path routing (MPR). Multiple identifiers can be assigned to each network node, and different routing paths between each distinct pair of identifiers can exist. MPR can provide a means of alleviating contention but not eliminating it. As an extension of this work, MPR could also be used to switch between multiple communication patterns that occur during application execution.



(a) $N_i -> N_i \pm 1$ (no contention)

(b) $N_i -> N_i \pm 2$ (with contention)

(c) $N_i -> N_i \pm 2$ (no contention)

Fig. 2. Example routing on a 16 node statically routed quaternary fat-tree network.

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3. Application communication patterns

Scientific applications commonly contain a limited number of communication activities such as boundary exchanges between logically adjacent processors, and collective operations. Most conform to the SPMD (Single-Program-Multiple-Data) style - that is each processor performs the same operation but on a different sub-grid of the overall global problem. When a communication activity occurs all processors typically perform it at the same time. The nature of the operation is the same for all processors. For boundary exchanges, the communication takes place between many processor-pairs

From recent analysis of scientific applications [13,14], the number of communication patterns is usually limited and the majority of applications exhibit regular communication patterns. Even when irregular patterns occur, they often contain regular patterns and can be mostly represented by a set of communication shifts [15]. Additionally the communicating processing-pairs, and hence the communication pattern, may be dynamic - it may change from one application iteration to the next but can persist for a reasonable amount of time before changing [13].

3.1. *Application topology*

A communication matrix is useful to examine the logical pattern of communication that occurs between processors within an application. This matrix is simply a two-dimensional representation of all possible communicating processor-pairs. One axis indicates the source, and the second axis indicates the destination. Four exam-

ple 128 processor communication patterns are shown in Figure 3. The vertical axis represents the sources of the communication (from 0 to $P-1$), and the horizontal axis represents the destination (again from 0 to $P-1$). A black pixel denotes a communicating processor-pair. Figure 3a) depicts a local-neighbor 2-D communication pattern for a 16×8 logical processor array with cyclic boundaries in the X dimension. Figure 3b) depicts a 3-D communication pattern for an $8 \times 4 \times 4$ logical processor array. Figure 3c) depicts an irregular communication pattern observed in an Adaptive Mesh Refinement (AMR) code.

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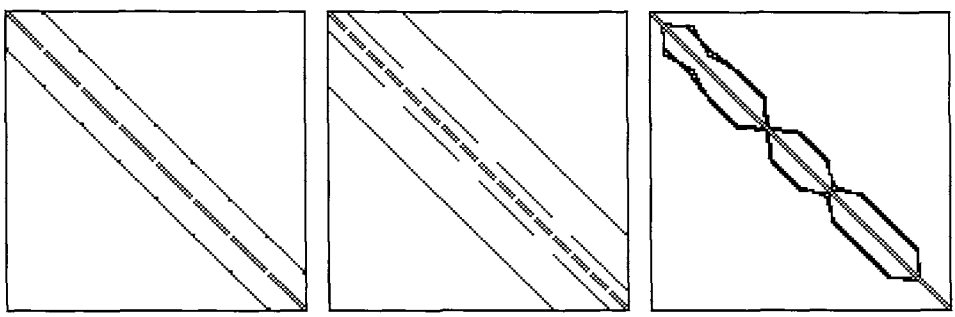

(a) 2-D cyclic in X (b) 3-D non-cyclic (c) Irregular

Fig. 3. Example communication patterns.

The communication matrix does not indicate the network components that are actually used, or the network topology. Rather it depicts the communication topology of the application (sometimes referred to as the virtual topology) and results from the way in which sub-grids interact. It is interesting to note however, that the 2-D and 3-D communication patterns of Figure 3 can be represented by a set of 1-D communications in which each processor sends to its neighbor processor a distance n apart. This can be considered as a communication *shift* distance. For example, in the case of the 2-D pattern, the shift distances are ± 1 (for communications in the logical X dimension - a 1-D pattern), and ± 16 (for communications in the logical Y dimension).

Another feature of the communication matrix is the occurrence of symmetry about the major diagonal. If this exists, as in Figure 3, then the communication is usually bi-directional, i.e. if processor P_i sends to processor P_j, then P_j also sends to P_i. A communication matrix can also be used to examine several characteristics of the communication pattern. Figure 3 uses only a binary value to denote whether a processor-pair communicated or not. Other quantities that may be of interest include the traffic between processors, message count, and average message sizes.

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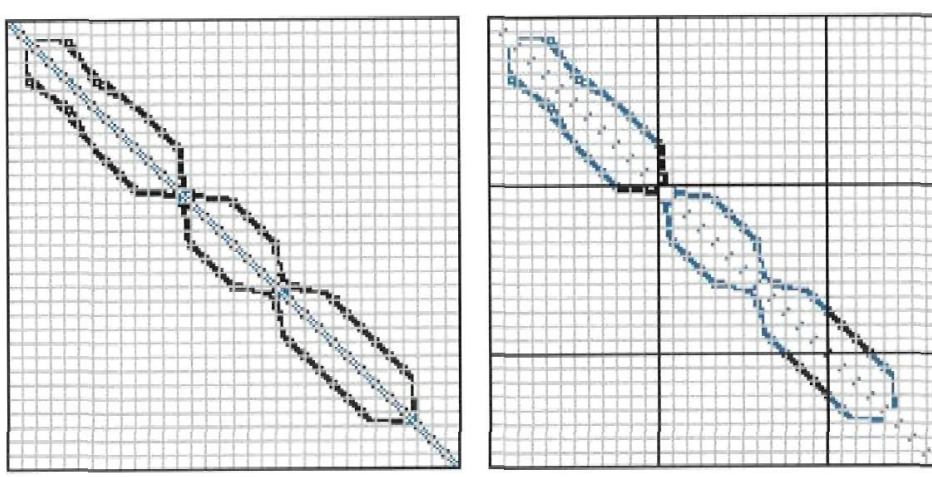
3.2. *Physical communication topology*

The layout of application tasks onto processors in a system and the way in which messages are routed determines the physical communication topology. Communications that occur between processors within the same compute-node need to be distinguished from those that occur between processors on different nodes. Intra-node communications typically use the nodes internal network, such as HyperTransport or shared memory, and the external HPC network is used for inter-node communications. As an example an irregular communication pattern is shown in Figure 4(a). The vertical axis represents the sources of the communication (from 0 to $P-1$), and the horizontal axis represents the destination (again from 0 to $P-1$). Intra-node communications are shown as grey pixels and inter-node communications are shown as black pixels. In this example, each node contains 4 processors and the extent of intra-node communications are shown by the faint horizontal and vertical lines.

The communications occurring within the first level of Infiniband switches is shown in Figure 4(b) for the example communication pattern. In a similar way to the intra-node example, the intra-switch communications are shown in grey, and the communications requiring the next level of switches are shown in black. The extent of each switch (48 PEs = 12 nodes x 4 PEs per node) is shown by the black horizontal and vertical lines. The communications that occur between nodes on the same switch can be handled simultaneously for different communicating nodes. Note that the bottleneck in this case is simply the number of messages originating from the same node since they share a single network NIC.



(a) 4-way nodes
(grey = intra-node, black = inter-node)

(b) first level of 12-port switches
(grey = intra-switch, black = inter-switch)

Fig. 4. An example 128 processor communication pattern.

An important aspect of this work is in the efficient utilization of the routes

between any two level-1 switches using switches in level-2. Even though in Figure 4(b) it appears as though there are only a few messages between the 1st and 2nd switch (as well as between the 2nd and 3rd switch), if these are routed via the same switch in an upper level of the network, they will effectively be serialized and share the same route down to the destination node. However, given the fat-tree nature of the Infiniband network, multiple routes are available but, due to the static routing employed in each of the network switches, they are not necessarily employed for a given communication pattern.

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3.3. *Communication routing and contention*

For a particular communication pattern the physical channels used within the network can be determined by analyzing the LFTs (routing tables) within each of the switches in the network. Since the routing is static, we can precisely determine the path taken by each message and hence determine the maximum contention that will occur for a particular pattern. The maximum contention will determine the performance of a particular communication stage within an application. For simplicity we have assumed that all messages within a communication pattern are of the same size in the following analysis. This assumption can be relaxed if necessary.

We use a routing array, $\mathbf{RA^m}$, in a similar way to [16] for each of the messages, m, contained within a communication pattern:

$$\mathbf{RA^m} = \begin{bmatrix} P_1^{1,1} & P_2^{1,1} & \dots & P_{24}^{1,1} & \| \dots \| & P_1^{1,n} & P_2^{1,n} & \dots & P_{24}^{1,n} \\ & & \vdots & & & & \vdots & & \\ P_1^{L,1} & P_2^{L,1} & \dots & P_{24}^{L,1} & \| \dots \| & P_1^{L,n} & P_2^{L,n} & \dots & P_{24}^{L,n} \end{bmatrix} \tag{1}$$

Each element of $\mathbf{RA^m}$, $P_x^{z,y}$, represents a port x on switch y at communication level z in the network. Individual switches are separated by the double vertical lines in Eq.(1). The number of communication levels in this description is twice the number of switch levels - one for each communication direction (both up to and down from a switch level). For instance the communication links from the compute nodes to the first level switches are represented by $z = 1$, and from the first level switches to the compute nodes are $z = 2$. For a fully-populated 288-port switch with 288 compute-nodes, the RA^m has 288 columns and 4 rows. $P_x^{z,y}$ is a binary number and is set to 1 if message, m, uses the particular channel in the network.

The contention array (**CA**) contains the number of messages that use each of the communication channels in the network for a particular communication pattern, and is the sum of the **RA**s for all messages, N_{msg}, within a communication pattern:

$$\mathbf{CA} = \sum_{i=1}^{N_{msg}} \mathbf{RA^i} \tag{2}$$

The maximum contention, $MaxC$, for a particular communication pattern is simply the largest value within **CA**. $MaxC$ determines the time for a particular

communication pattern in an SPMD application and can be thought of as a multiplicative factor on the messaging time for large messages. $MaxC$ can be minimized by optimizing the routing of all messages within a communication pattern.

Note that the calculation of the maximum contention is just one use of the **CA**. For instance the number of unused links, the distribution of contention, and the use of MPR can also be examined.

4. Infiniband routing table analysis

The routing tables within the Infiniband fabric can be stored to file when using OpenSM. In addition, it also enables a predefined map (set of routing tables) for the whole network to be loaded at any point. Part of a routing table is shown below in which the LID specifies a destination address in the system, the output port is specified by a value between 1 and 24, and the Destination Info contains GUIDs for the destination device. Switch output ports in the range 1 thru 12 are up-links in the network (to a switch in a higher level in the network), and ports in the range 13-24 are down-links which on the lowest switch level are connected to the compute nodes. LIDs in the range 5-10 for a single switch are shown below. Note that devices in the network also include the switches themselves (e.g. LID 8 below).

LID	Output Port	Destination Info
0x0005	004	(Adapter portguid 0x0008f1040396fa01: 'MT25208')
0x0006	005	(Adapter portguid 0x0008f1040396fb01: 'MT25208')
0x0007	006	(Adapter portguid 0x0008f1040396fc01: 'MT25208')
0x0008	010	(Switch portguid 0x0008f10400400905: 'ISR9288')
0x0009	008	(Adapter portguid 0x0008f1040396f405: 'MT25208')
0x000a	009	(Adapter portguid 0x0008f1040396f505: 'MT25208')

A typical routing map was obtained using the OpenSM's dump facility. In addition, two custom routing maps were defined and are described in Section 4.1 below, which provide an optimized set of routing tables as well as a worst-case set of routing tables for a 2-level, 288-port, Infiniband cluster. The contention that arises from these maps, as well as from the typical map, is compared in Section 4.2 for a number of communication patterns.

4.1. *Routing table definition*

In the following we consider one particular communication pattern - that of a logical shift: $P_i -> P_i \pm s$ where s is the shift distance and ranges from 1 to 128. As described in Section 3 this particular set of communication patterns can be considered a superset of other patterns including those occurring from 1-D, 2-D and 3-D regular data decomposition schemes. Three separate network maps are used which we refer to as Typical, Worst-case, and Optimized:

- **Typical** - a map built by the OpenSM subnet manager after a reboot, and that represents a typical case calculated using the default UP/DOWN routing method.

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- **Worst-case** - a map in which a routing table for each lowest level switch was manually defined that routes messages destined for all compute-nodes attached to any other switch through a single up- link. The maximum contention in this case is equal to the number of compute-processors attached to a switch (48 in our test system).
- **Optimized** - a map in which a routing table for each lowest level switch was manually defined that allows 12 messages (one per up-link) to be routed simultaneously to any other lowest level switch. Thus, this routing table best enables a logical shift communication pattern of any shift distance to be embedded in the network with no contention.

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For a two level network, as shown in Figure 1, and considering the level-1 switches to be number L1 thru L24, the worst-case routing tables are defined by:

```
For each Level 1 switch n = 1 .. 24
    For all Nodes not connected to Ln use output port n/2
```

and Optimized routing tables are defined by:

```
For each Level 1 switch n = 1..24
    For all Nodes not connected to Ln use output port Node%12
```

As a system scales in size, the performance of a typical map will approach that of the worst-case. For all three maps the routing down-wards in the network is deterministic as there is at most one route that can be taken. No alteration was made to the routing tables in these cases. An optimized map for networks with a greater number of switch levels could be similarly defined in which each higher level of switches spreads out the communication traffic for the shift communication pattern across their up-links.

4.2. *Contention calculated from the routing tables*

Routing arrays, as described in Section 3.3, were used to calculate the contention array for a particular communication pattern. For each inter-node message in the communication pattern, a separate routing array, $\mathbf{RA^m}$, is constructed using the routing tables from the switches through which a message is routed. The contention array is calculated by the summing all routing arrays as given by Eq.(2). The result of this procedure provides a count of the number of messages that are routed across each network link. The maximum represents the longest time a message within the pattern will take to be communicated.

In Figure 5, the maximum contention as calculated by Eq.(2) is shown for the logical shift communication pattern on the full test system of 256 nodes for each of the three maps. The shift distance is varied from 1 to 128 processors. The characteristics of the curves in Figure 5 illustrate the full range of contention possible within the network.

First it should be noted that the worst-case map plateaus at a contention of 48 whereas the optimized map plateaus at a contention of only 4. Recall that the worst-case map routes all messages from the processors within its local switch that are destined for a compute-node attached to another switch though a single up-link - a maximum of all 48 local processors. The shift distance actually determines the number of messages that will be routed through an up-link - a shift distance of 1 results in only one message routed from one level-1 switch to another - this number gradually increases linearly up to the maximum of 48.

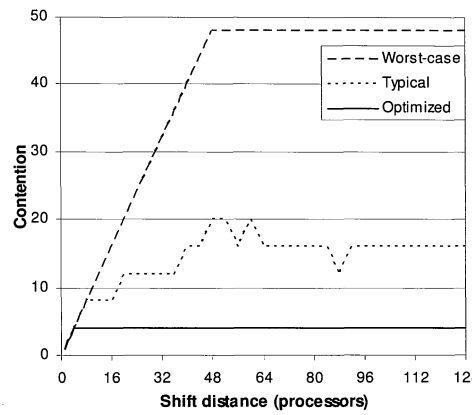

Fig. 5. Maximum contention resulting from the Infiniband routing tables (logical-shift pattern on 1024-processors)

In the case of the optimized map, the contention increases linearly to 4 and then plateaus. The height of this plateau is simply the size of a compute-node in terms of processing cores (assuming a single MPI process per core). A maximum of 4 is reached when all 4 processors perform an inter-node communication (at a shift distance of 4 or greater). Since the optimized map allows all 12 up-links to be simultaneously used to communication from one level-1 switch to another, no further contention occurs (i.e. the optimized map has 1/12 the contention of the worst-case map at shift-distances of 48 or greater).

The use of a typical map results in a contention factor that is between the worst-case and the best-case, and that generally increases with shift-distance. Though the contention level may also decrease as shown in Figure 5 as a result from the precise routing tables employed and the communication pattern (or shift distance in this case) used. In the experience that we have gained from our test-system, the typical case is closer to that of the optimized case - though we expect the typical case to get worse as a system scales in size. The effect of scale is considered further in Figure 6 below when using smaller segments of our test-system.

The contention that arises from 256 processor jobs and 512 processor jobs when using the logical shift communication pattern is shown in Figure 6. The contention that occurs in both the worst-case and optimized maps is the same as for the 1024

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processor job (as shown in Figure 5) - and hence only the optimized case is included in Figure 6. The minimum, maximum and average contention is calculated in both Figure 6(a) and 6(b). This range occurs since both a 256 or 512 processor can be mapped to multiple sets of nodes the 256 node test system and hence make different use of the network. The base node for these tests was altered from 0 up to 192 (256 processor job) and from 0 up to 256 (512 processor job) in increments of 4 nodes. The maximum contention that occurs is the same for job sizes of 1024, 512, and 256 processors. Also note that the minimum increases towards the maximum with increasing job size.

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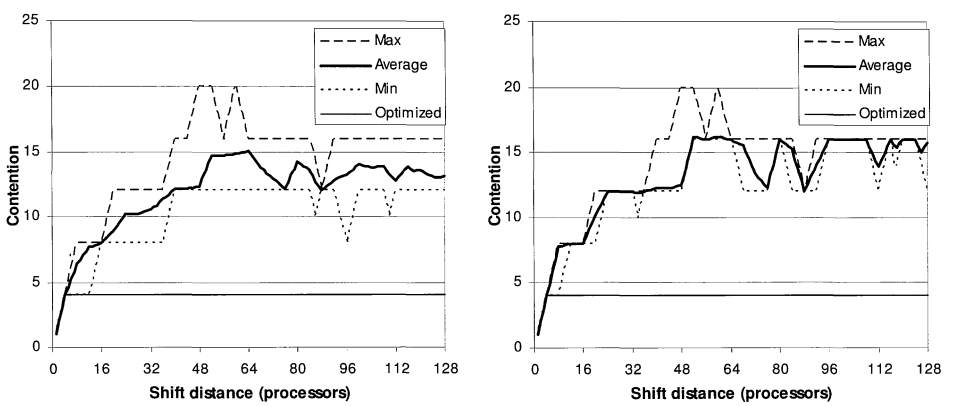

(a) 256 processor job (1-D pattern) (b) 512 processor job (1-D pattern)

Fig. 6. Contention calculated from IB routing tables for the logical-shift pattern when varying base-node position.

A similar analysis was performed for two further communication patterns - a 2-D and 3-D pattern. In the 2-D pattern the processors are arranged in a P_x by P_y ($P_x \times P_y = P$) logical array and communications occur in the X dimension : $P_i -> P_i \pm 1$, and in the Y dimension $P_i -> P_i \pm P_x$ (ignoring edge effects). In the 3-D pattern processors are arranged in a P_x by P_y by P_z ($P_x \times P_y \times P_z = P$) logical array and communications occur in the X dimension $P_i -> P_i \pm 1$, in the Y dimension $P_i -> P_i \pm P_x$, and in the Z dimension $P_i -> P_i \pm P_x \times P_y$.

Note that the communication in any of the logical dimensions in these two communication patterns are very similar to a logical shift. For instance, the communication in the X dimension is a shift of 1 with the communications from boundary processors removed, and similarly the communication in the Y dimension is a shift of P_x again with boundaries removed.

The contention that is observed using the **CA** for communications in the Y dimension and Z dimension is shown in Figure 7 (contention in the X dimension is always 1). The contention arising from the typical, worst-case and optimized maps can be seen for different processor counts (corresponding to different logical

processor array sizes) from 8 up to 1024 processors. The contention for the 2-D and 3-D patterns correspond almost directly to that observed for the logical shift pattern in Figure 5. For instance, consider the contention of the Z communication in the $8 \times 8 \times 8$ 3-D processor array which is 16, 48, and 4 for the three maps as shown in Figure 7. The Z communication in this case is similar to a logical shift of distance $8 \times 8 = 64$. As seen in Figure 5, the contention for this logical shift is the same at 16, 48, and 4 for the three maps respectively.

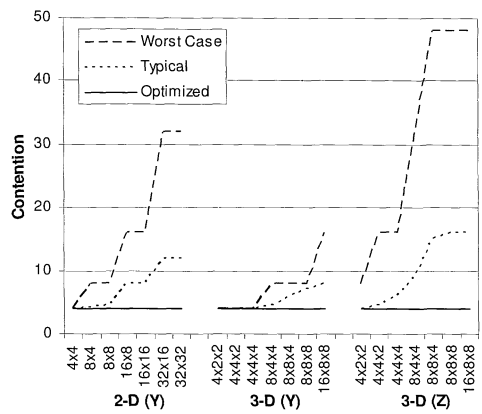

Fig. 7. Contention calculated from IB routing tables for 2-D and 3-D communication patterns.

5. Measurement of communication pattern performance

Here we use a communication micro-benchmark that implements the logical shift communication pattern and is parameterized in terms of message size and processor count. The micro-benchmark was implemented using the coNCePTuaL network correctness and performance testing utility [17]. The pseudo code of the benchmark is:

```
Barrier, Time T1
For n repetitions do
    ISend to self + distance,a message of S Bytes
    IRecv from self +- distance, a message of S Bytes
    Waitall
Barrier, Time T2
```

where distance is the logical shift distance. This benchmark was used to obtain the time taken to perform the logical shift communication pattern for $S = 1$MBytes for the three maps. Figure 8 shows the time taken (in ms on the right-hand Y-axis) for the logical shift communication pattern for a shift distance up to 128. Also shown is the contention using the routing tables presented in Section 3. The scale of the Y-axis was deliberately chosen to examine the correspondence between the measured

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time and the modeled contention. Exact correspondence can be seen for the typical map, very good correspondence for the optimum case, and exact correspondence up to a shift distance of 48 in the worst-case.

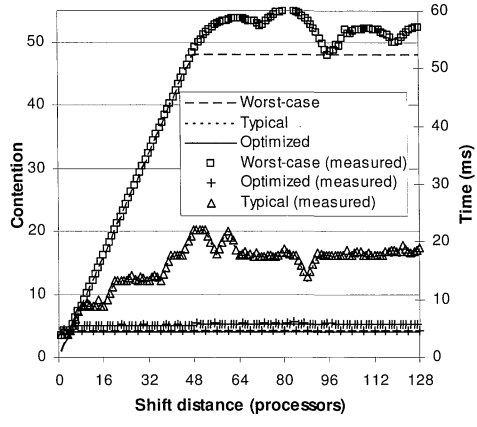

Fig. 8. Comparison of modeled contention (left Y-axis) and measured communication time (right Y-axis) for a 1024-processor job

The impact of message size on the actual communication time is also of concern. In Figure 9 we examine the communication time for messages between 8KBytes and 1MBytes for the three maps. The metric of communication time per byte is used in Figure 9 (or 1/bandwidth) so as to provide curves in term of the time taken that are comparable across message sizes. It can be seen that the time per byte varies to the greatest extent in the case of the optimized map (by almost a factor of 2) from the small messages to the large messages. A far smaller range in time is observed in the case of the typical and worst-case maps. This is due to the increased contention level in both of the latter two maps. When the contention level is low, as in the case of the optimized map, there is less chance for small messages to actually collide in the network, even when they are routed though the same channel. As the contention level increases, the chance of collision is much greater and hence the communication time per byte follows the contention as calculated by the routing tables.

6. Impact on application performance

In this section we examine the effect of using different maps on application performance. Two large-scale applications, SAGE and Partisn are used in this analysis. SAGE is an adaptive mesh hydrodynamics application that is used for the simulation of shock-waves. The performance characteristics of SAGE have been studied and modeled in some detail [18]. Partisn is an implementation of SN transport, the solution of the Boltzmann equation using the discrete ordinates method, on structured meshes. Some details on Partisn can be found in [19]. Both applications consist of many hundred thousand lines of fortran90, and can run on thousands of

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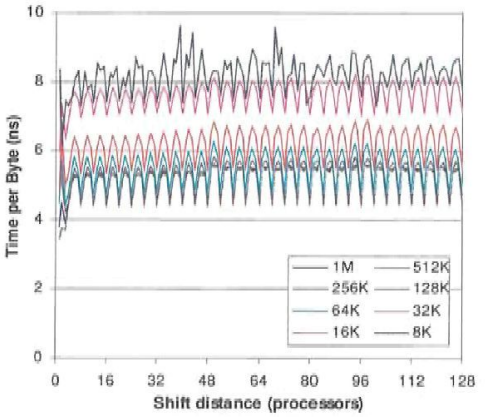
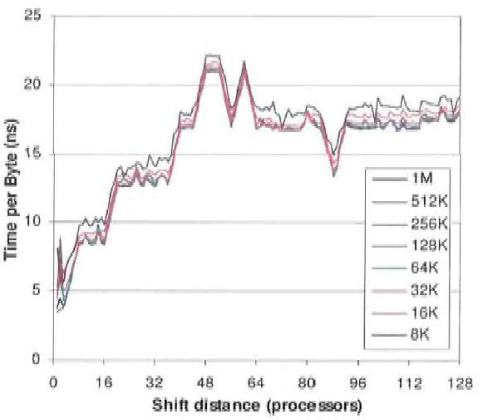
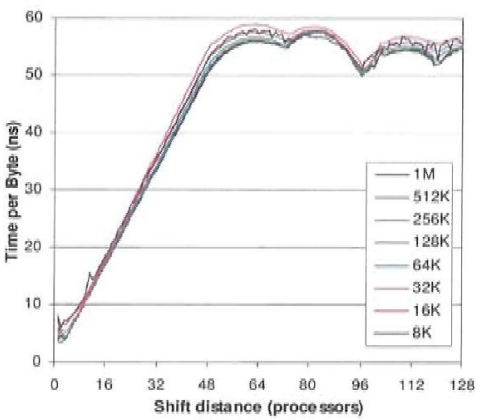


(a) Optimized map



(b) Typical map



(c) Worst-case map

Fig. 9. Communication performance (time per byte) for various message sizes using the worst-case, typical and optimized routing tables

processors for long periods of time. They are iterative and many iteration cycles are undertaken for a physical simulation. A summary of the characteristics of both applications as well as the input decks used in this analysis is listed in Table 1.

The input used for SAGE contains a 2-D global data grid which is partitioned in only 1 dimension. A 3-D problem is used in Partisn and is partitioned in two dimensions. Both applications are run in weak-scaling mode in which parallelism is used to increase the resolution of the simulation. In weak-scaling mode, the run-time (or time per cycle) is ideally a constant but in general increases with scale due to parallel overheads including communication costs.

SAGE exhibits a logical shift communication pattern (described earlier). The shift distance gradually increases from 1, at small scale, up to a distance of 24 at 1024 processors for the input deck considered. The 2-D data decomposition used in Partisn results in a shift distance of 1 for communications in the logical X dimension, and a shift distance of P_x in the logical Y dimension. The dimensions of the logical 2-D processors array ($P_x \times P_y$) in Partisn is chosen to be close to a square and thus varies between 1 and 32 when using up to 1024 processors.

Table 1. Application Characteristics.

	SAGE	Partisn
Input Deck	Rbeq_5	Rep1
Problem calculation	2-D	3-D
Scaling	Weak	Weak
Sub-grid size (cells/PE)	12,800	13,500
Logical topology	1-D (+)	2-D mesh
Cycle time on 1 node (s)	0.7	60
Message count /PE/cycle	4K	600K
Message sizes (Bytes)	8, 50K	8, 1K, 15K
Collectives	Allreduce Alltoall Reduction	Allreduce Broadcast

The performance of both applications was measured using the typical, worst-case and optimized maps. The measured cycle time for SAGE is shown in Figure 10(a) and for Partisn is shown in Figure 10(b). It can be seen that the highest level of performance (lowest time) is achieved when using the optimized map, worse performance is achieved when using the typical map, and substantially worse performance is achieved when using the worst-case map. The actual improvement in performance, from the typical map to the optimized map, increases with scale and is up to 16% for SAGE and 8% for Partisn. We expect that this difference will be even greater for larger systems due to the bulk synchronous behavior of the applications, if one MPI task is delayed due to contention, all ranks wait for it at the next collective operation.

It should be noted however that the network map optimization only affects a part of the total execution time. In Figure 11 we show the relative performance of just the communication cost of the typical and worst-case maps relative to the

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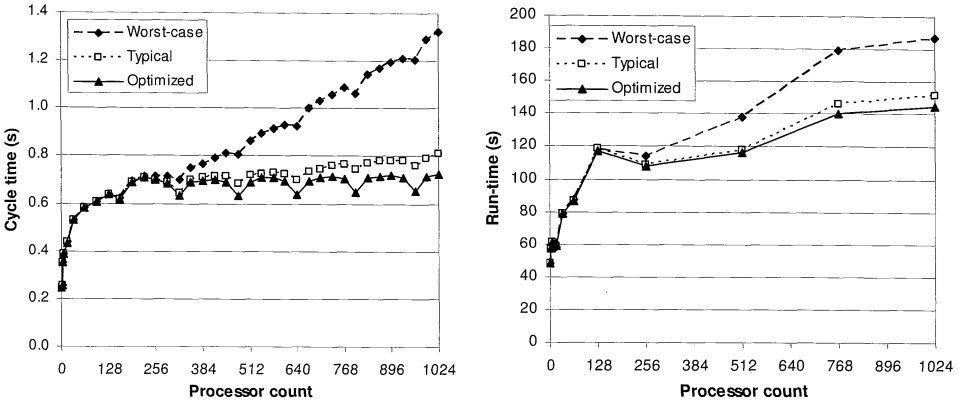

(a) SAGE (b) Partisn

Fig. 10. Application performance using worst-case, typical, and optimized routing tables.

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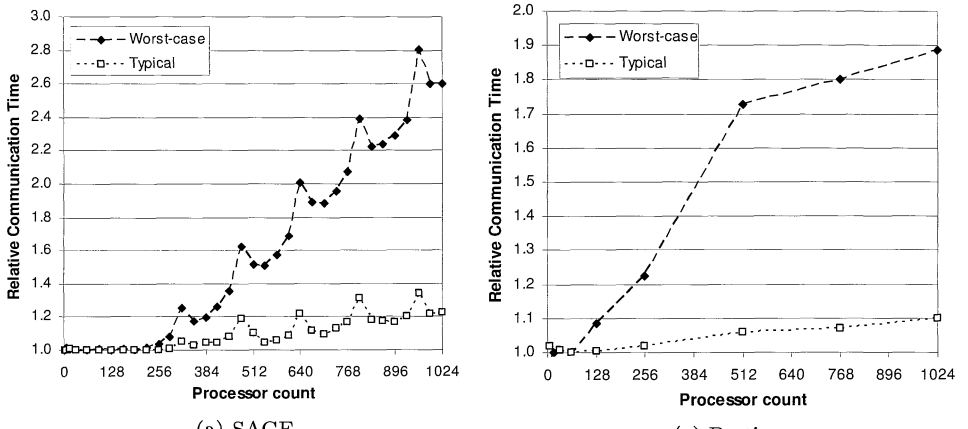
optimized map. A value greater than one includes a lower performance, for example a relative performance of 1.2 indicates that the communication cost is 20% longer.

It can be seen in Figure 11 that the communication performance when using the optimized map is up to 40% faster in SAGE, and 10% faster for Partisn. The difference is substantially greater when considering the relative communication time between the worst-case and the optimized map - up to 2.8× on SAGE, and 1.9× on Partisn. SAGE has a communication distance of up to 24 at the largest scale for its major communications whereas Partisn has a distance of up to 32 for only half of its communications (in only one of its two decomposition dimensions) and in addition many of its communications are small. This to a great extent explains the higher improvement observed for SAGE than for Partisn.



(a) SAGE (a) Partisn

Fig. 11. Communication time relative to optimum when using worst-case and typical tables.

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7. Conclusions

In this work we have quantified the effect of static routing in a large-scale Infiniband cluster for a range of communication patterns. Contention can impact communication performance by up to a factor of 12 higher than in an optimized case. Two custom routing tables for the network were defined: a worst-case map in which only one up-link from a switch is used to route messages, and an optimized map which uses all 12 up-links from a switch simultaneously for a logical shift communication pattern.

Through an analysis of the routing tables for a particular communication pattern it was shown how the maximum contention within the network could be calculated. Further, a set of micro-benchmarks confirmed that this maximum contention did indeed determine the time for a communication pattern. Two large-scale applications of interest to Los Alamos showed that the optimization of the routing tables resulted in improvements in application performance. Overall the run-time on the applications decreased by between 10-20% at the largest scale analyzed (1024 processors on 256 compute-nodes). Further the communication component of the run-time decreased by between 20-40% at the largest scale. We expect the performance improvements to be even greater at larger scale. The optimization of the routing tables presented in this work has applicability to many currently installed Infiniband systems.

From these results further research into adaptive routing for Infiniband is warranted. Barriers to this work include the time it takes to load these routes to switches (several seconds). Possible solutions depend on the persistence of the communication patterns within the applications and also what portion of this work if any could be moved into the HCA or switches.

Acknowledgements

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